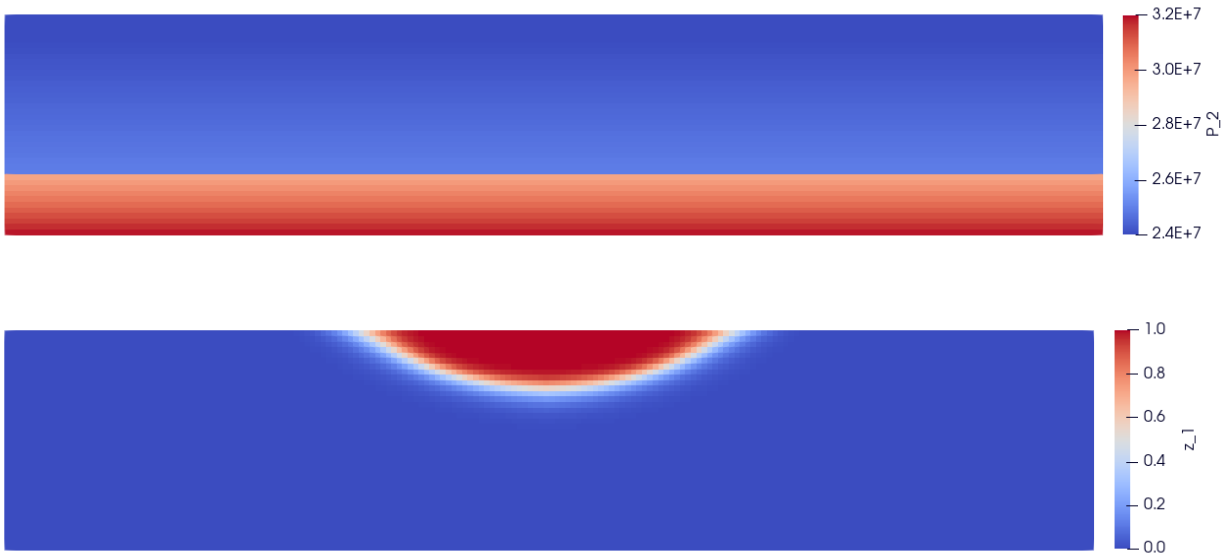
DARSim Example:

UHS in depleted gas field (H2-CH4-Brine simulation)



This example demonstrates how to model underground hydrogen storage (UHS) in a depleted gas field using the H_2–CH_4–brine branch. The reservoir is 800 m thick: the upper 600 m is initially saturated with CH_4, while the lower 200 m is saturated with brine. The figures show the initial reservoir pressure and the H_2 mass fraction after 200 days of hydrogen injection.

To configure the fluid model and component list, specify the following:

```
FLUID_MODEL
      Compositional
      --Number of phases
      2
      --Number of components
      3

COMPONENT_LIST
  H2-CH4-Brine
```

The component indices in this system are: 1: H_2 , 2: CH_4 , 3: Brine

The initial conditions can be specified in a text file using the INITIAL_STATE keyword:

```
INITIAL_STATE
      Initial_State.txt
```

The text file should contain the following columns: cell index, liquid-phase pressure (P2), H_2 mass fraction (z1), CH_4 mass fraction (z2), and brine mass fraction (z3).

Diffusion and dispersion can be specified as follows:

```
DIFFUSION_CONSTANTS
      1E-7
      1E-9

DISPERSIVITY
      1E-2
```

Finally, the injection and production schedule, together with the injected component, can be specified inside the well block:

```
SCHEDULE
      50, 250
COMPOSITION
      1
```

In this example, the well injects component 1 (H_2) from day 50 to day 250. The schedule can be extended to define additional well-opening and shut-in periods.

Brine salinity can be specified in [ppm] using the SALINITY keyword:

```
SALINITY
      1E4
```

Gas-phase viscosity is calculated internally as a function of temperature, pressure, and gas composition. Therefore, its value can be set to Nan. The liquid-phase viscosity, however, must be specified explicitly:

```
FLUID_VISCOSITY (Pa.sec)
      Nan
      5.5E-4
```

The first entry corresponds to the gas-phase viscosity, while the second specifies the liquid-phase viscosity.